Eyebrow Measurement Under Head Movement

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Abstract

Head movement can distort estimated eyebrow position and complicate the measurement of expression. Two experiments examined neutral-distance prediction and expression preservation. The OpenFace experiment used 4,414 neutral frames from 63 recordings and eight signers. A multilayer perceptron (MLP) had lower prediction error than ridge regression when training and validation shared recordings, but ridge performed better when complete recordings were held out. Stronger regularization brought the MLP closer to ridge in an exploratory follow-up without establishing an overall advantage. A separate Face Landmarker pilot used 22 recordings of one participant to evaluate pitch correction. Relative to each recording's frontal reference height, a generalized additive model (GAM) reduced mean neutral root mean squared error (RMSE) by 26.9% and increased held-raised RMSE by 29.8%. Correction also reduced separation between neutral and raised states, showing that improved neutral stability did not ensure expression preservation in this pilot.

1 Introduction

Eyebrow movement contributes to meaning in sign languages, and automated measurements can help describe its timing and shape. Head movement, however, can distort the estimated position of the eyebrows. A correction must therefore account for pose-related distortion while retaining the expression being studied.

Previous studies have addressed this measurement problem using facial landmark tracking and learned correction. Kuznetsova et al. (2022) corrected OpenFace measurements in a study of question marking in Kazakh-Russian Sign Language. Kuznetsova and Kimmelman (2024) subsequently compared OpenFace with MediaPipe Holistic and identified difficulties in measuring eyebrow movement during head tilt. Building on this work, the OpenFace experiment examines how validation design affects model comparisons, while the separate Face Landmarker pilot tests whether pitch correction preserves eyebrow raises.

The basic measurement is the perpendicular distance from an eyebrow point q to the line through two eye corners, a and b, given by

$$d(q, a, b) = \frac{\|(q - a) \times (b - a)\|}{\|b - a\|}. \tag{1}$$

Rigid rotation or translation does not change this distance for exact three-dimensional coordinates. Variation during a fixed expression may therefore reflect tracking or reconstruction errors, or movement that is not fully rigid.

The OpenFace correction model predicts neutral eyebrow distance from head pose and identity inputs. Subtracting that prediction from the observed distance produces a residual signal whose usefulness depends on both prediction accuracy and retention of real eyebrow changes.

2 OpenFace experiment

2.1 Data and models

The experiment used processed OpenFace measurements released by Kuznetsova et al. (2022). OpenFace estimates facial landmarks and head pose from video (Baltrušaitis et al., 2018).

The selected neutral subset contained 4,414 frames from 63 recordings, eight signers, and ten sentences. All frames were retained, including 65 with tracking confidence below 0.8. Section 2.1 of the source study reports replacing low-confidence measurements with neighbouring values. Retaining confidence flags therefore does not establish that the released coordinates were unchanged.

The prediction targets were the released inner- and outer-brow distances in their original coordinate units, without conversion to millimetres. Inputs comprised three head rotations in radians, their cosines, three translations, and indicators for signer and sentence identity. Category encoding and input standardization were fitted on the training data only, and the targets remained unscaled.

Ridge regression and an MLP were compared with baselines that predicted the overall training mean or each signer's training mean. An additional ridge control used signer and sentence indicators without pose to assess the contribution of pose inputs.

The fixed ridge penalty was 0.001. The MLP had one hidden layer with 40 rectified linear units and two outputs, using Adam with a penalty of 0.0001 and a maximum of 1,500 iterations without automatic early stopping. Seed 42 defined the main fit, with seeds 7 and 19 used to assess sensitivity to initialization.

2.2 Validation

Random frame validation assessed prediction within familiar recordings. Four splits each held out 25% of the frames, with every recording represented in both training and validation. Validation frames could recur across splits.

Recording-level validation assessed prediction on complete recordings excluded from training. Four folds evaluated each of the 63 recordings once. All validation signers and sentences also appeared in training, but their combinations were new. Each signer–sentence combination corresponded to one recording, so the design could not distinguish recording novelty from combination novelty.

Performance was summarized by joint mean squared error (MSE), averaging the squared errors of the two eyebrow targets. Frame-split scores weighted frames equally and were averaged over the four splits. Recording-fold scores were calculated within each recording and then averaged with equal weight for all 63 recordings. Training used equal frame weights in both designs. The different scoring weights prevent interpreting the difference between the two summaries as a direct percentage effect of holding out recordings.

2.3 Results

Table 1. Joint MSE for the fixed OpenFace comparison. Frame splits use equal frame weights and share recordings between training and validation. Recording folds use equal recording weights and hold out complete recordings. Lower values indicate better prediction, and MLP results use seed 42.

Model	Frame splits	Recording folds
Overall mean	17.958	16.058
Signer mean	12.073	11.311
Ridge with all inputs	1.318	2.471
Ridge with identity only	—	12.601
MLP, seed 42	0.429	31.797

The MLP outperformed ridge on random frame splits but had substantially higher error when complete recordings were held out (Table 1). At seed 42, it performed worse than ridge on 58 of the 63 recordings. Seeds 7 and 19 showed the same pattern, with recording MSEs of 40.850 and 40.203. Ridge retained its advantage when the recording predictions were scored with equal frame weights, giving MSE 2.467 for ridge and 36.253 for the main MLP.

Pose inputs improved ridge's neutral-distance predictions. The control using identity indicators alone had recording MSE 12.601, compared with 2.471 for the model that also included pose.

3 Exploratory MLP follow-up

3.1 Model selection

The follow-up examined whether stronger regularization and simpler input sets could improve the MLP's performance on held-out recordings. It was designed after inspection of the original results and reused the same data and outer folds. The three input sets were pose with signer and sentence identity, pose with signer identity, and pose alone.

Model selection used three recording-grouped inner folds within each outer training set. The MLP penalties were 0.0001, 1, and 100, while ridge used 0.001, 1, and 100. The penalty for each input set and the MLP input set were selected by joint MSE with equal recording weights, using seed 42 for the MLP fits. Ridge was tuned through the same grouped inner validation.

3.2 Results

Inner validation selected a penalty of 100 for every MLP input set in every outer fold and consistently chose pose with signer identity.

Table 2. Exploratory joint MSE on held-out recordings, with equal recording weights and seed 42 for the MLP fits. Fixed ridge provides the original comparison.

Model	Joint MSE
Fixed ridge	2.471
Ridge with grouped penalty selection	2.600
MLP with pose, signer, and sentence	3.106
MLP with pose and signer	2.623
MLP with pose only	6.093

Stronger regularization reduced the full-input MLP's recording MSE from 31.797 to 3.106. The selected model using pose and signer inputs reached MSE 2.623, compared with 2.600 for tuned ridge and 2.471 for fixed ridge (Table 2). Its score ranged from 2.597 to 2.623 across seeds 7, 19, and 42, and at seed 42 it outperformed fixed ridge on 27 of 63 recordings.

Recording-level prediction offsets dominated the original MLP's errors, accounting for 95.6% of its MSE with equal recording weights at seed 42. Checks of responses to signer and sentence inputs were consistent with overfitting to their combinations, although extreme translation values may also have affected fitting. Restricting scoring to frames with confidence at least 0.8 still left ridge well ahead of the original MLP.

4 Face Landmarker pilot

4.1 Recordings and measurement

The pilot examined pitch, or upward and downward head tilt, using head-movement videos from Kuznetsova and Kimmelman (2024). The sample comprised 22 recordings of one participant at close, middle, and far camera distances. Eleven recordings had neutral eyebrows, six had a held raise, and five alternated between neutral and raised eyebrows. One additional alternating clip was excluded because it lacked a usable frontal reference.

Landmarks were extracted with MediaPipe Face Landmarker in video mode, using MediaPipe version 0.10.32 and model bundle `float16/1`. A single face was tracked with a fresh tracker for each recording. The task returns estimated three-dimensional landmarks (Google, 2026) and differs from the source study's MediaPipe Holistic pipeline. Earlier facial geometry work describes approximate mesh reconstruction from monocular video (Kartynnik et al., 2019).

Each recording used a stable frontal video segment as its reference. I selected these segments before viewing correction results, using 0.20–0.50 seconds with at least five valid frames. The reference could contain neutral or raised eyebrows and established the recording's eyebrow level, eye-span scale, and reference pose.

Normalized eyebrow height was calculated from coordinates adjusted for image dimensions. The landmark x and z coordinates were multiplied by image width, and y by image height. The mean perpendicular distance from inner-brow landmarks 55 and 285 to the line through eye corners 133 and 362 was divided by the median distance between eye landmarks 33 and 263 in the reference segment. This dimensionless measure differs in scale from the OpenFace targets, preventing direct comparison of errors between the experiments.

Relative pitch was derived from the facial transformation matrix after removing scale through polar decomposition. Each frame's rotation relative to the mean reference rotation was calculated as $R_{\text{ref}}^{\mathsf{T}} R_{\text{frame}}$, and pitch was extracted in degrees using extrinsic xyz Euler angles. Positive pitch denoted downward tilt, verified against visible movement. Missing detections and invalid geometry remained missing without interpolation.

Visible eyebrow states provided labels for evaluating expression preservation in the alternating clips. I labelled clear neutral and raised intervals, excluding transitions and uncertain moments. The annotations identified nine clear raise episodes.

4.2 Correction and validation

The correction models predicted deviation from each neutral recording's median reference height using pitch. Both pitch and target values were standardized using training data only. Each neutral training recording contributed an equal number of evenly spaced valid frames, capped at 150 and limited by the training recording with the fewest valid frames.

The GAM fitted a penalized cubic spline with the basis parameter set to six and penalties of 0.1, 1, and 10. The MLP used eight tanh hidden units, L-BFGS, penalties of 0.001, 0.01, and 0.1, and a maximum of 3,000 iterations without automatic early stopping. Seed 42 defined the main fit, with seeds 7 and 19 used to check sensitivity to initialization. Candidates that failed to converge were excluded from selection.

The correction subtracts the predicted pose-related deviation while leaving the measurement unchanged at zero reference pitch. For raw normalized height h and predicted deviation $g(p)$ at relative pitch p, it is defined as

$$h^{\text{corr}} = h - \left[g(p) - g(0) \right]. \tag{2}$$

Models were evaluated by holding out one camera distance at a time. Only neutral recordings from the other two distances contributed to fitting and tuning. Inner validation trained at one distance and validated at the other, then reversed these roles. The penalty with the lowest mean recording RMSE within the training pitch range was selected, and the model was refitted on both training distances. Evaluation covered all eyebrow conditions at the held-out distance, with each recording using its own frontal reference.

The primary comparison used identical valid frames for every method, excluding reference segments and pitch values outside the training range. A separate complete-clip comparison retained raw measurements outside that range.

For neutral and held-raised recordings, RMSE quantified stability relative to the recording's raw median reference height. A held raise should remain near its raised reference level during head movement, so this measure assessed whether correction increased or reduced variation within that state.

For alternating recordings, area under the receiver operating characteristic curve (AUROC) quantified separation of the labelled states, with higher height always treated as raised. AUROC of 1 indicates perfect separation, while a constant height gives 0.5 when both states are present. Both RMSE and AUROC were calculated separately for each recording and averaged with equal recording weights. Percentage changes used unrounded recording averages.

4.3 Results

Table 3. Pilot results within the training pitch range, averaged equally across recordings. RMSE is expressed in normalized eye-span units for 11 neutral and six held-raised recordings. AUROC uses five alternating recordings. MLP results use seed 42. Lower RMSE and higher AUROC indicate better performance. The public pooled summary gives unrounded means and verified recording counts.

Method	Neutral RMSE	Raised RMSE	Alternating AUROC
Raw	0.01329	0.02000	0.9575
GAM	0.00972	0.02595	0.9000
MLP	0.00999	0.02681	0.9000

Correction improved neutral stability but worsened held-raised stability (Table 3). Mean neutral RMSE fell by 26.9% for the GAM and 24.9% for the MLP. For held-raised recordings, it increased by 29.8% and 34.1%, respectively.

Separation of the alternating states also declined, with mean AUROC falling from 0.9575 to 0.9000 for both models. This change was confined to one close-distance recording, whose AUROC dropped from 0.7875 to 0.5. The other four recordings retained AUROC of 1.0.

The complete-clip comparison retained the same pattern. Keeping raw measurements outside the training pitch range gave a GAM reduction in neutral RMSE of 26.7% and an increase in held-raised RMSE of 17.1%. The MLP's additional seeds also gave similar results.

5 Discussion and limitations

Neutral-distance prediction alone provides an incomplete basis for evaluating correction. The OpenFace MLP performed well within familiar recordings but poorly on new ones, while the pilot models improved neutral stability at the expense of held-raised stability and state separation. Evaluation for use on new recordings therefore needs to test both transfer between recordings and retention of expression.

The OpenFace results concern new recordings of known signers and sentences. Each held-out signer–sentence combination also corresponded to a new recording, so the effects of a new combination and a new recording cannot be separated.

The follow-up reused the original held-out recordings after their results had informed the model changes, so its scores should be interpreted as exploratory. A penalty of 100 was selected for every MLP input set, leaving penalties above 100 untested.

The pilot used one participant, which limits conclusions about performance across people. Errors in selecting a frontal reference or assigning a visible-state label could affect the scores. Missing detections and pitch values outside the training range reduced the frames available for the primary comparison.

Both experiments evaluated video-based estimates of eyebrow position. The OpenFace frames labelled neutral may still contain small eyebrow movements or tracking errors. In the pilot, Face Landmarker provided both pose and eyebrow position, so their estimation errors may be related.

6 Conclusion

Ridge gave the lowest error in the fixed OpenFace comparison on complete held-out recordings, and grouped tuning improved the MLP without establishing an overall advantage. In the one-participant pilot, pitch correction improved neutral stability but worsened held-raised stability and reduced state separation. Further evaluation should test correction on new participants and recordings, with head movements covering a wider pitch range.

Code and data availability

The project repository contains code, settings, and saved results (Al-Jomaa, 2026), including source URLs and hashes for retrieving the OpenFace inputs. Participant-linked pilot files, such as annotations and individual recording metrics, are excluded from the public release. Exact pilot reproduction requires the original local inputs.

References

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